WORKSHEET

Using expected values

1 A discrete random variable X takes values in the set $S = \{2,3,4,\ldots,12\}$ and has probability function

$$P(X = x) = \begin{cases} \frac{x-1}{36} \text{ for } 2 \leq x \leq 7 \\ \frac{13-x}{36} \text{ for } 8 \leq x \leq 12 \end{cases}$$

Determine:

a $P(X = 5)$

b $P(X = 7)$

c $P(X = 11)$

d $E(X)$

2 A game costs \$3 to play where you have a 0.3 chance of winning \$5 and a 0.7 chance of winning \$2. What is your expected return?

3 What is the expected value of rolling a standard six-sided die?

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4 Two dice are rolled. What is the expected value of the difference of the two dice?

5 Two dice are rolled. What is the expected value of the product of the multiplication of the two dice?

6 Find the expected value given the following probability distributions.

a

x	\$10	\$20	\$50	\$100
$p(x)$	0.4	0.3	0.2	0.1

b

y	100	1000	10 000	100 000
$p(y)$	0.889	0.1	0.01	0.001

c

x	1	2	3	4	50
$p(x)$	0.24	0.25	0.25	0.25	0.01

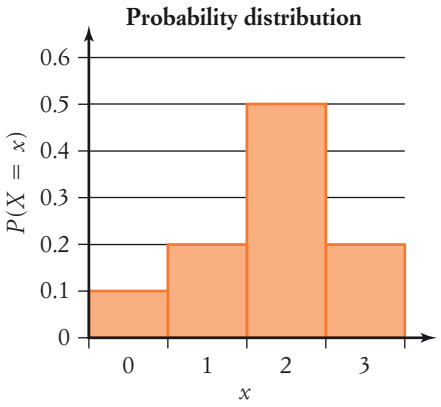
7 A standard deck of 52 cards is used to randomly draw a single card. If it is a picture card (K, Q, J) you pay me \$3, and if it is not a picture card I will pay you \$1. What is your expected return?

8 You hit the bullseye on a dart board 5% of the time. What is your expected return if you receive \$50 for hitting the bullseye and nothing for missing?

9 A bag contains three cards numbered 1 and five cards numbered 3. What is the expected value when a card is randomly chosen?

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10 Find the expected value given the probability distribution of discrete random variable X below.



11 A standard die is rolled a maximum of five times or until a 6 appears. If X = the number of rolls of the die, find the expected value of X.

Answers

1 **a** $\frac{1}{9}$

b $\frac{1}{6}$

c $\frac{1}{18}$

d 7

2 Lose \$0.10

3 3.5

4 ≈ 1.944

5 12.25

6 **a** \$30

b 388.9

c 2.99

7 Win \$0.08

8 \$2.50

9 2.25

10 1.8

11 ≈ 1.579

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